Programming Robots ⚙️

Cadette Robotics Badge — Girls Build Bots

Presented by your robotics mentors!

The Control Loop

Every robot on Earth runs the same fundamental cycle:



A robot is **a program running in a loop, connected to the physical world.**

Trace it on a real robot 🔍

Robot vacuum: - **SENSE:** bump sensor pressed, cliff sensor sees stairs - **THINK:** “obstacle ahead → pick a new direction” - **ACT:** motors reverse and turn - **…and repeat, ~50 times per second**

Your turn: trace the loop for a self-driving car. What could go wrong if the loop runs too slowly?

Today’s programming toolkit 🧰

Concept	What it does	Example
Event	reacts to a trigger	on sprite overlap
Loop	repeats behavior	forever: move
Conditional	makes decisions	if wall hit → turn
Variable	remembers a value	packages = 3

These four ideas power everything from games to Mars rovers.

Build: Autonomous Warehouse Robot 🏭

In MakeCode Arcade:

1. Robot sprite in a warehouse map with walls
2. **Sense:** detect wall collisions
3. **Think:** if wall hit → change direction; track packages in a **variable**
4. **Act:** move continuously in a *forever* loop
5. Collect package sprites on overlap

The Autonomy Challenge 🎯

Make your robot collect ALL packages with ZERO keyboard input.

Pure programmed behavior. No driving.

This is the difference between *operating* a machine and *engineering* one.

Stretch goals: - 🔋 Battery variable that drains + charging station - 🤖🤖 Second robot — avoid collisions!

Code Review: How real teams work 👥

Swap laptops with a partner. On your review sheet:

1. What does this program do?
2. Point to a **conditional**. Point to a **variable**.
3. One thing you'd improve.

Then discuss — and improve your own code with their feedback.

Every professional software team on Earth does exactly this.

What you accomplished ✅

- Explained the sense→think→act control loop
- Programmed autonomous behavior — no human input
- Used events, loops, conditionals, and variables
- Ran a real code review

This is genuinely how robots are programmed. FRC robots run the same logic — just in a bigger language.

Level up 🚀

- **arcade.makecode.com** — finish your stretch goals
- **microbit.org** — program real hardware (~$20)
- **python.org / Codecademy** — text-based coding is your next step
- **FIRST Tech Challenge** (firstinspires.org) — program real competition robots in Java
- Ask us what programming an FRC robot is like!